Configured BMR sign control passed

The first two studies fixed the model structure. This third study instead checks the sign of a configured Bayesian-model-reduction comparison. Its estimand is the BMR free-energy difference for two candidate likelihood-column prunings.

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Configured BMR sign... (part 2)

The categorical diagnostic is related to the model-reduction mechanism discussed by Friston et al. (Friston et al. 2024) and the post-hoc BMR lineage (Friston and Penny 2011). It is not a discovery claim about structure emergence.

Configured BMR sign... (part 3)

The fixed posterior ranges over $n = 4$ candidate states. One likelihood column is unsupported by the configured evidence; another is a supported control. BMR substitutes a reduced prior for each candidate and computes ΔF eq. 20. This is one deterministic closed-form comparison,

Configured BMR sign... (part 4)

so no resampled sample, confidence interval, or paired test applies (Smith et al. 2022).

Configured BMR sign... (part 5)

ΔF is positive for the declared redundant-column pruning, so the reduced candidate is favored for this fixed posterior. **Redundant-column result:** $\Delta F = 3.68$; the configured reduction is accepted.

Configured BMR sign... (part 6)

It is negative for the declared supported-column control. **Supported-column control:** $\Delta F = -27.67$; that reduction is rejected.

Configured sign-control disposition: redundant accepted and supported rejected, recorded as Yes.

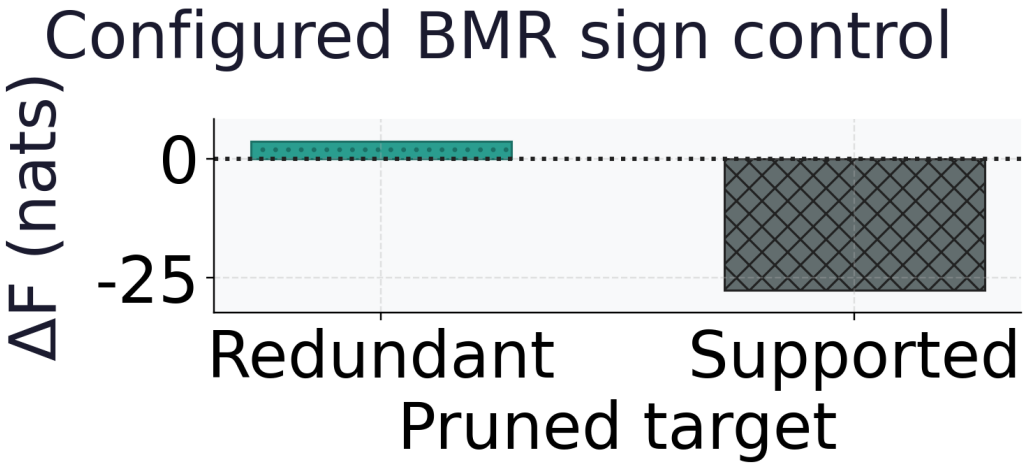
Configured BMR sign... (part 7)

The observed sign pattern $\Delta F_{\text{redundant}} > 0 > \Delta F_{\text{supported}}$ passes the configured control: this fixed posterior favors pruning the declared redundant column and rejects pruning the declared supported column.

Configured BMR sign... (part 8)

It does not show that arbitrary data or model families will discover, prune, or retain the correct structure.

Configured BMR sign… (part 9)



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Configured BMR sign... (part 10)

Redundant target ΔF: 3.677 nats.
Supported target ΔF: -27.67 nats.
Positive gain favors the reduction;
negative gain rejects it.

Configured BMR sign... (part 11)

Source configured sign-control verdict: True. This is not an optimizer convergence diagnostic.

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Configured BMR sign... (part 12)

One fixed-posterior categorical diagnostic. No independent stochastic replication or uncertainty interval applies. Source-mechanism analogue, not full protocol reconstruction.

Configured BMR sign... (part 13)

fig. 15 contrasts the two configured prunings and reports the sign control.

Configured BMR sign… (part 14)

Studies 1–3 all ran in a *trusting* world, where every broadcast belief is honest. The contamination sweep that follows removes that assumption, and it is the point at which the three robustness axes of sec. 10.2.1 begin to diverge.